

Deep Double Descent

Where bigger models, longer training, and more data can hurt

COMP34312 | Mathematical Topics in Machine Learning | Poster Coursework

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Objective Explain why test error can peak *near interpolation* and then fall again.
Problem Classical bias–variance theory predicts one U-shape; modern overparameterised nets interpolate yet generalise.
Approach Sweep width, training time, samples, noise, optimiser, regularisation; track test *and* train error.
Contribution Nakkiran et al. unify model-wise, epoch-wise, and sample-wise double descent via **Effective Model Complexity** (EMC).

1. From the textbook U-curve to the double-descent peak

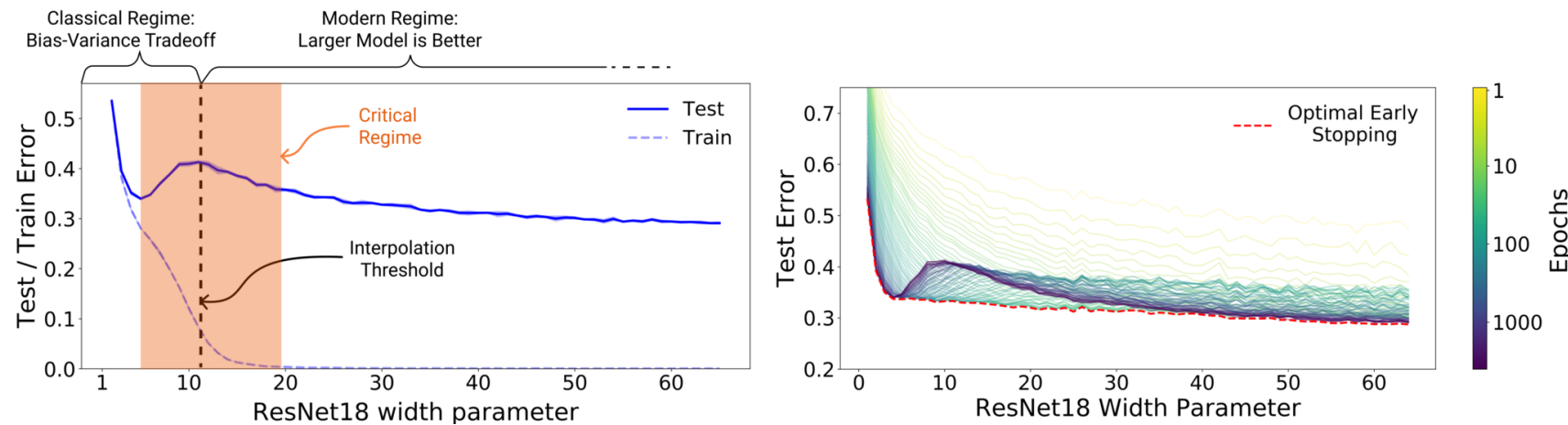


Fig. 1, Nakkiran et al. (2019). Left: test error on ResNet18s, varying width, CIFAR-10 + 15% label noise. Right: test error coloured by training epoch; dashed red curve denotes optimal early stopping.

2. Setting: the lecture decomposition we extend

Following the course notation (Notes §5.1, Eqs. 68 to 73):

$$\hat{f} \leftarrow \mathcal{A}(\mathcal{F}; S), \quad R(f) := \mathbb{E}_{\mathbf{xy}}[\ell(\mathbf{y}, f(\mathbf{x}))], \quad \hat{R}(f, S) := \frac{1}{n} \sum_{i=1}^n \ell(\mathbf{y}_i, f(\mathbf{x}_i)).$$

The excess risk decomposes into three statistically meaningful gaps:

$$\underbrace{R(\hat{f}) - R(y^*)}_{\text{excess risk}} = \underbrace{R(\hat{f}) - R(f_{\text{erm}})}_{\text{optimisation}} + \underbrace{R(f_{\text{erm}}) - R(f^*)}_{\text{estimation}} + \underbrace{R(f^*) - R(y^*)}_{\text{approximation}}.$$

The classical bias-variance trade-off bundles *approximation + part of estimation* into **bias** and the rest into **variance**; what the lecture summary calls *incomplete*, not wrong.

- Lecture U-curve** (Notes Fig. 17) holds $\mathcal{F}, \mathcal{A}$ fixed and grows $\mathcal{F}$.
- Paper's pivot**: treat the *whole procedure* $\mathcal{T} = (\mathcal{F}, \mathcal{A}, \text{epochs, reg., labels})$ as the single object whose capacity matters, so optimisation, estimation, and labels jointly determine the peak.

3. Definition + Hypothesis (Nakkiran et al. 2019)

DEFINITION 1: EFFECTIVE MODEL COMPLEXITY

A **training procedure** $\mathcal{T}$ maps any labelled train set $S = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^n$ to a classifier $\mathcal{T}(S)$. For an error tolerance $\varepsilon > 0$,

$$\text{EMC}_{\mathcal{D}, \varepsilon}(\mathcal{T}) := \max\left\{n \mid \mathbb{E}_{S \sim \mathcal{D}^n}[\text{Error}_S(\mathcal{T}(S))] \leq \varepsilon\right\}$$

i.e. the largest sample size on which $\mathcal{T}$ can still drive train error below ε . Note that EMC depends on the optimiser, training time, augmentation and labels, not just on the architecture.

HYPOTHESIS 1: GENERALISED DOUBLE DESCENT

For “natural” $\mathcal{D}$, neural-network procedures $\mathcal{T}$, and small ε (the paper uses $\varepsilon = 0.1$):

EMC $\ll n$

Under-parameterised: any change to $\mathcal{T}$ that *increases* EMC *decreases* test error.

EMC $\approx n$

Critically parameterised: the sign of the same change is *indeterminate*.

EMC $\gg n$

Over-parameterised: any change to $\mathcal{T}$ that *increases* EMC *decreases* test error.

Why this is a hypothesis, not a theorem. ε and the words “sufficiently smaller / larger” are heuristic; the critical-interval width is not formally specified for deep nets. **EMC is operational, not predictive**: there is no closed form, so the peak location must be *measured*, not derived.

4. Main empirical evidence: model-wise $\times$ epoch-wise double descent

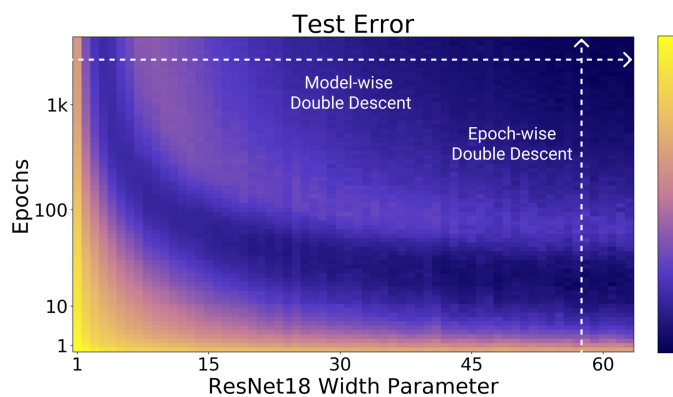


Fig. 2 (left), Nakkiran et al. — test error. Test error attains ≈ 0.47 at width $k \approx 10$ and decreases to ≈ 0.24 at $k = 64$.

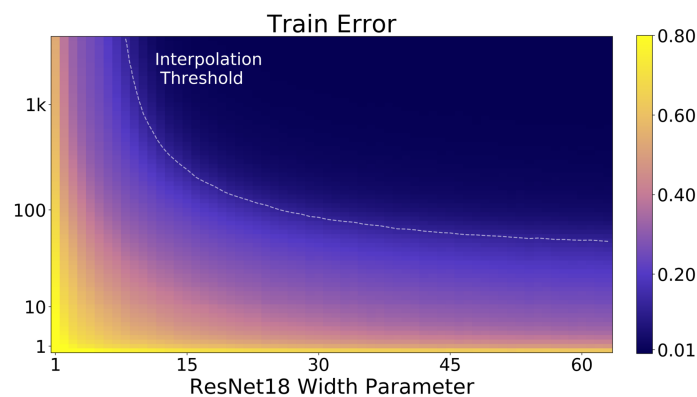


Fig. 2 (right), train error. Dashed curve denotes the interpolation threshold ($\hat{R} \rightarrow 0$).

READING THE HEATMAPS

- Same axes**: ResNet18 width $k \in [1, 64]$ vs training epochs (log).
- Horizontal slice**: **model-wise** peak as width grows. **Vertical slice**: **epoch-wise** peak as training continues.
- Key alignment**: the bright *test* ridge sits on the *train* interpolation curve, so the peak is *at* the threshold.

SETUP + KEY NUMBERS (PAPER §4)

ResNet18, CIFAR-10, **15% label noise**, augmentation, Adam (lr 1e-4), 4k epochs. Pattern identical under SGD, on CIFAR-100, and at **0% noise** (Fig. 4a). Sample-wise (Fig. 3): a **4.5 $\times$ data increase** (4k $\rightarrow$ 18k) raises test loss for $d_{\text{model}} \in [50, 100]$. Paper uses $\varepsilon = 0.1$ in EMC.

The unifying variable is EMC, not parameter count: any modification of $\mathcal{T}$ that moves EMC across n shifts the peak correspondingly.

5. Sample-wise descent: increasing n can locally raise test risk

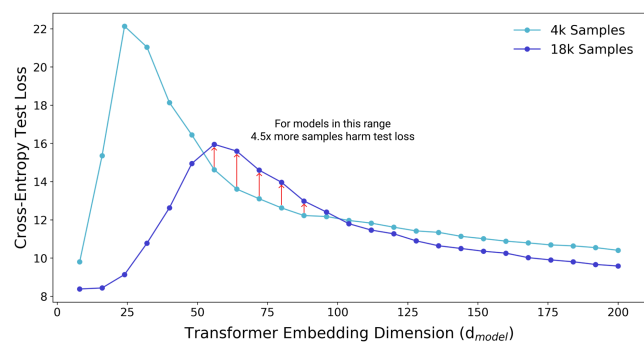


Fig. 3, Nakkiran et al. IWSLT'14 De-En, 6-layer Transformer, $d_{\text{ff}} = 4d_{\text{model}}$, 80k gradient steps, 10% label smoothing. Highlighted bars indicate the d_{model} interval over which a $4.5\times$ increase in n raises test loss.

- Fixing $(\mathcal{T}, \mathcal{F})$ and increasing n leaves the interpolation threshold in EMC space unchanged but shifts n relative to it; widths near the small-sample threshold are thereby displaced into the critical regime.
- The non-monotonicity is local: asymptotically, additional samples reduce excess risk.

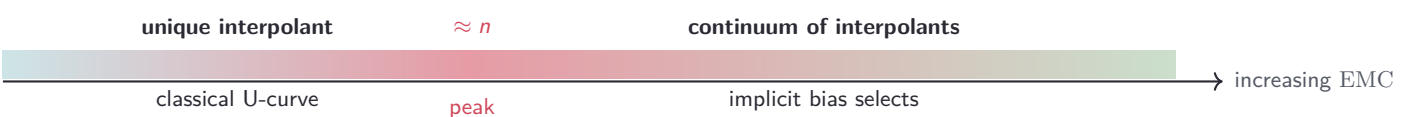
6. Mechanism and implication chain (rigorous for linear models; conjectural for deep nets)

CRITICALLY PARAMETERISED

The hypothesis class admits effectively a unique interpolant. Constraining it to fit noisy or mis-specified labels perturbs the on-distribution map; test risk consequently spikes.

OVER-PARAMETERISED

A continuum of interpolants exists. The optimiser's **implicit bias** (e.g. SGD toward minimum-norm solutions) selects an interpolant that absorbs label noise while preserving smoothness on the data manifold.



WHERE THE MECHANISM IS PROVEN

Mei & Montanari (2022, Thm. 2) establish a test-risk peak at $d=n$ for random-features regression under mis-specification; Belkin et al. (2019) prove an analogous peak for least-squares; Bartlett et al. (2020) prove benign overfitting under minimum-norm interpolation. The deep-network analogue remains conjectural.

IMPLICATION CHAIN

Decreasing n at fixed $\mathcal{T}$ increases EMC/ n , shifting the regime further into over-parameterisation; an **ensemble of K models trained on disjoint sub-samples** therefore lies past each member's peak, and averaging cancels member-specific interpolation noise (Nakkiran et al., Fig. 28).

8. Practical implications and open questions

IMPLICATIONS FOR PRACTICE

- Diagnostic**. Train and test curves decouple near the interpolation threshold; monitor both.
- Per-model mitigation**. Weight decay, early stopping, or scaling well beyond the threshold.
- Aggregate mitigation**. Ensemble K sub-sampled models, each over-parameterised relative to its own n (Fig. 28).
- Augmentation**. Data augmentation shifts the peak rightward (Fig. 5) without changing the architecture.

OPEN SCIENTIFIC QUESTIONS

- A principled choice of ε and closed-form critical-interval width.
- A non-linear EMC theorem for over-parameterised networks.
- Characterisation of SGD's implicit bias for realistic architectures and losses.

Interpolation is a threshold to understand, not a final verdict.

KEY REFERENCES

[1] Nakkiran, Kaplan, Bansal, Yang, Barak, Sutskever. *Deep Double Descent: Where Bigger Models and More Data Hurt*. arXiv:1912.02292 (2019). [2] Belkin, Hsu, Ma, Mandal. *Reconciling modern ML practice and the classical bias-variance trade-off*. PNAS 116(32) (2019). [3] Zhang, Bengio, Hardt, Recht, Vinyals. *Understanding deep learning requires rethinking generalization*. ICLR (2017). [4] Mei & Montanari. *The generalization error of random features regression*. Comm. Pure Appl. Math. (2022). [5] Bartlett, Long, Lugosi, Tsigler. *Benign overfitting in linear regression*. PNAS 117(48) (2020). [6] Brown & Mukherjee. *COMP34312 Notes 2026, §5*.

AI-USE DECLARATION

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